

# Sepehr Hajebi – CV

August 2026

## CONTACT

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## APPOINTMENTS

- **University of Toronto**, Department of Mathematics 2026 – present  
Postdoctoral Fellow holding a Canada Postdoctoral Research Award
- **University of Waterloo**, Combinatorics and Optimization 2024 – 2026  
Postdoctoral Scholar
- **Princeton University**, Department of Mathematics 2024  
Junior Faculty (Instructor)  
*Not taken up due to U.S. entry restriction against all Iranian nationals*

## EDUCATION

- **University of Waterloo**, PhD in Combinatorics and Optimization 2020 – 2024  
Thesis: Foreshadowing the Grid Theorem for Induced Subgraphs  
Advisor: Sophie Spirkl
- **Isfahan University of Technology**, BSc and MSc in Mathematics 2013 – 2019

## AWARDS

- **Canada Postdoctoral Research Award**, 2026 (140,000 CAD)  
Natural Sciences and Engineering Research Council of Canada (NSERC)
- **Graduate Research Excellence Award**, 2025 (2500 CAD)  
University of Waterloo
- **Mathematics Doctoral Prize: 1<sup>st</sup> place**, 2025 (1500 CAD)  
University of Waterloo
- **Governor General's Gold Medal**, Finalist Designation, 2025  
University of Waterloo
- **Sinclair Graduate Scholarship**, 2023-2024 (4000 CAD)  
University of Waterloo
- **Outstanding TA Award**, 2022  
University of Waterloo
- **Math competitions**: *International Mathematics Competition for University Students* (Bronze Medals – 2015 and 2016), *Korean Mathematical Contest* (Bronze Medal – 2016), *Iranian Mathematical Competition* (Silver and Bronze Medals – 2015 and 2016)
- **Elite Student Recognition**, 2016 and 2018  
Isfahan University of Technology

## PUBLICATIONS

### ► JOURNAL PAPERS

- [1] *Induced subgraphs and tree decompositions XIX. Thetas and forests*  
**SIAM Journal on Discrete Mathematics** 40 (2026), 1000–1009.  
*with M. Chudnovsky, J. Codsí and S. Spirkl.*
- [2] *Tree independence number III. Thetas, prisms and stars*  
**Journal of Graph Theory** 106 (2024), 923–943.  
*with M. Chudnovsky and N. Trotignon.*
- [3] *Suns in triangle-free graphs of large chromatic number*  
**Electronic Journal of Combinatorics** 33 (2026)  
*with S. Spirkl.*
- [4] *Induced subgraphs and tree decompositions XVI. Complete bipartite induced minors*  
**Journal of Combinatorial Theory, Series B** 176 (2026), 287–318.  
*with M. Chudnovsky and S. Spirkl.*
- [5] *Tree independence number II. Three path configurations*  
**Journal of Combinatorial Theory, Series B** 176 (2026), 74–96.  
*with M. Chudnovsky, D. Lokshtanov and S. Spirkl.*
- [6] *Chordal graphs, even-hole-free graphs and sparse obstructions to bounded treewidth*  
**Journal of Graph Theory** (2025), 1–15.
- [7] *Induced subgraphs and tree decompositions IX. Grid theorem for perforated graphs*  
**Advances in Combinatorics** 2025:3, 40pp.  
*with B. Alecu, M. Chudnovsky and S. Spirkl.*
- [8] *Induced subgraphs and tree decompositions XII. Grid theorem for pinched graphs*  
**Innovations in Graph Theory** 2 (2025), 1–23.  
*with B. Alecu, M. Chudnovsky and S. Spirkl.*
- [9] *Induced subgraphs and tree decompositions XIV. Non-adjacent neighbors in a hole*  
**European Journal of Combinatorics** 124 (2025), 104–074.  
*with M. Chudnovsky and S. Spirkl.*
- [10] *Induced subdivisions with pinned branch vertices*  
**European Journal of Combinatorics** 124 (2025), 104–072.
- [11] *Induced subgraphs and tree decompositions VI. Graphs with 2-cutsets*  
**Discrete Mathematics** 348 (2025), 114–195.  
*with T. Abrishami, M. Chudnovsky and S. Spirkl.*
- [12] *Induced subgraphs and tree decompositions XIII. Basic obstruction in  $\mathcal{H}$ -free graphs for finite  $\mathcal{H}$*   
**Advances in Combinatorics** 2024:6, 30pp.  
*with B. Alecu, M. Chudnovsky and S. Spirkl.*
- [13] *List- $k$ -Coloring  $H$ -free graphs for all  $k \geq 4$*   
**Combinatorica** 44 (2024). 1063–1068.  
*with M. Chudnovsky and S. Spirkl.*
- [14] *Induced subgraphs and tree decompositions VIII: Excluding a forest in (theta, prism)-free graphs*  
**Combinatorica** 44 (2024), 921–948.  
*with T. Abrishami, B. Alecu, M. Chudnovsky and S. Spirkl.*
- [15] *Tree independence number I. (Even hole, diamond, pyramid)-free graphs*  
**Journal of Graph Theory** 106 (2024), 923–943.  
*with T. Abrishami, B. Alecu, M. Chudnovsky, S. Spirkl and K. Vůskovič.*

- [16] *List-3-Coloring ordered graphs with a forbidden induced subgraphs*  
**SIAM Journal on Discrete Mathematics** 38 (2024), 1158–1190.  
*with Y. Li and S. Spirkl.*
- [17] *Hitting all maximum stable sets in  $P_5$ -free graphs*  
**Journal of Combinatorial Theory, Series B** 165 (2024), 142–163.  
*with Y. Li and S. Spirkl.*
- [18] *Induced subgraphs and tree decompositions VII. Basic obstructions in  $H$ -free graphs*  
**Journal of Combinatorial Theory, Series B** 164 (2024), 443–472.  
*with T. Abrishami, B. Alecu, M. Chudnovsky and S. Spirkl.*
- [19] *Induced subgraphs and tree decompositions II. Toward walls and their line graphs in graphs of bounded degree*  
**Journal of Combinatorial Theory, Series B** 164 (2024), 371–403.  
*with T. Abrishami, M. Chudnovsky, C. Dibek, P. Rzażewski, S. Spirkl and K. Vůskovič.*
- [20] *Induced subgraphs and tree decompositions V. One neighbor in a hole*  
**Journal of Graph Theory** 105 (2023), 542–561.  
*with T. Abrishami, B. Alecu, M. Chudnovsky, S. Spirkl and K. Vůskovič.*
- [21] *Induced subgraphs and tree decompositions IV. (Even hole, diamond, pyramid)-free graphs*  
**Electronic Journal of Combinatorics** 30 (2023)  
*with T. Abrishami, M. Chudnovsky and S. Spirkl.*
- [22] *Induced subgraphs and tree decompositions III. Three-path-configurations and logarithmic treewidth*  
**Advances in Combinatorics** 2022: 6, 29 pp.  
*with T. Abrishami, M. Chudnovsky and S. Spirkl.*
- [23] *Complexity dichotomy for List-5-Coloring with a forbidden induced subgraph*  
**SIAM Journal on Discrete Mathematics** 36 (2022), 2004–2027.  
*with Y. Li and S. Spirkl.*
- [24] *Minimal induced subgraphs of two classes of 2-connected non-Hamiltonian graphs*  
**Discrete Mathematics** 345 (2022).  
*with J. Cheriyan, Z. Qu and S. Spirkl.*
- [25] *Edge clique cover of claw-free graphs*  
**Journal of Graph Theory** 90 (2019), 311–405.  
*with R. Javadi.*

► PREPRINTS

- [26] *Forcing monochromatic induced subgraphs*  
**arXiv:2606.24695**  
*with S. Spirkl.*
- [27] *Asymmetric induced saturation*  
**2606.24763**  
*with X. Fan, Sahab Hajebi and S. Spirkl.*
- [28] *Tree-alpha and excluding finitely many graphs*  
**arXiv:2605.01223**  
*with S. Spirkl.*
- [29] *Polynomial bounds for pathwidth*  
**arXiv:2510.19120.**

- [30] *A simple layered-wheel-like construction*  
arXiv:2410.16495  
with M. Chudnovsky, D. Fischer, S. Spirkl and B. Walczak.
- [31] *Halfway to induced saturation for even cycles*  
arXiv:2505.24100  
with X. Fan, Sahab Hajebi and S. Spirkl.
- [32] *Bull-free graphs and  $\chi$ -boundedness*  
arXiv:2504.21093.
- [33] *Induced subgraphs and tree decompositions XVIII. Obstructions to bounded pathwidth*  
arXiv:2412.17756.  
with M. Chudnovsky and S. Spirkl.
- [34] *Induced subgraphs and tree decompositions XVII. Anticomplete sets of large treewidth*  
arXiv:2411.11842.  
with M. Chudnovsky and S. Spirkl.
- [35] *Tree independence number IV. Even-hole-free graphs*  
arXiv:2407.08927  
• Conference version at **SODA (2025)**, 4444–4461.  
with M. Chudnovsky, P. Gartland, D. Lokshtanov and S. Spirkl.
- [36] M. Chudnovsky, P. Gartland, S. Hajebi, D. Lokshtanov and S. Spirkl  
*Induced subgraphs and tree decompositions XV. Even-hole-free graphs have logarithmic treewidth*  
with M. Chudnovsky, D. Lokshtanov and S. Spirkl.
- [37] *Induced subgraphs and tree decompositions XI. Local structure in even-hole-free graphs of large treewidth*  
arXiv:2309.04390  
with B. Alecu, M. Chudnovsky and S. Spirkl.
- [38] *Induced subgraphs and tree decompositions X. Towards logarithmic treewidth for even-hole-free graphs*  
arXiv:2307.13684  
with T. Abrishami, B. Alecu, M. Chudnovsky and S. Spirkl.

## TALKS

- *Polynomial pathwidth* (2026)  
Barbados Graph Theory Workshop, Bellairs Research Institute in Holetown, Barbados.
- *Complete bipartite induced minors (and treewidth)* (2025)  
Willaim Tutte Colloquium, University of Waterloo.
- *Local and global structure of non-basic obstructions to bounded treewidth* (2025)  
CanADAM (invited at the Structural Graph Theory Minisymposium), University of Ottawa.
- *“Underview” of excluded induced subgraphs for bounded pathwidth* (2025)  
Barbados Graph Theory Workshop, Bellairs Research Institute, Holetown, Barbados.
- *The pathwidth theorem for induced subgraphs* (2025)  
IBS discrete math seminar, Daejeon, South Korea.
- *The pathwidth theorem for induced subgraphs* (2024)  
Graphs and Matroids Seminar, University of Waterloo.
- *Anticomplete sets of large treewidth* (2024)  
BIRS Workshop on New Perspectives in Coloring and Structure, Banff, Alberta.

- *Survey on induced subgraphs of graphs with large treewidth* (2024)  
Barbados Graph Theory Workshop, Bellairs Research Institute in Holetown, Barbados.
- *Treewidth, Erdős-Pósa and induced subgraphs* (2024)  
New York Combinatorics Seminar. [Virtual]
- *Several Gyárfás-Sumner-type results for treewidth* (2023)  
Graphs and Matroids Seminar, University of Waterloo.
- *Hitting all maximum stable sets in  $P_5$ -free graphs* (2023)  
Graphs and Matroids Seminar, University of Waterloo.
- *Forests in even-hole-free graphs of large treewidth* (2022)  
Barbados Graph Theory Workshop, Bellairs Research Institute in Holetown, Barbados.
- *Holes, hubs, and bounded treewidth* (2022)  
IBS Discrete Math Seminar. [Virtual]
- *Bounded treewidth in hereditary graph classes* (2022)  
Graphs and Matroids Seminar, University of Waterloo.
- *Bounded treewidth in hereditary graph classes* (2022)  
Seymour was 70 two years ago, ENS de Lyon, France.

## TEACHING

- **Teaching at the University of Toronto**
  - MATH 344 Introduction to Combinatorics (Fall 2026)
- **Teaching at the University of Waterloo**
  - CO 380 Mathematical Discovery and Invention (Spring 2026)
  - MATH 135 Algebra for Honours Mathematics (Winter 2026)
  - MATH 116 Calculus I for Engineering (Fall 2025)
- **TA work at the University of Waterloo**
  - CO 342 Graph Theory (Spring 2023)
  - MATH 138 Calculus II for Honors Mathematics (Winter 2023)
  - MATH 600 Mathematical Software (Fall 2022)
  - CO 456 Game Theory (Fall 2022 and 2023)
  - CO 380 Mathematical Discovery and Invention (Spring 2022)
  - MATH 239 Introduction to Combinatorics (Winter 2022)
  - CO 255 Advanced Optimization (Winter 2022)
  - CO 250 Introduction to Optimization (Fall 2021 & 2023, Winter 2023 & 2024, Spring 2024)
  - CO 442/642 Graph Theory (Fall 2021)
  - CO 351 Network-Flow Theory (Spring 2021)
- **TA work at Isfahan University of Technology**
  - Computational Complexity (2019)
  - Elements of Matrices and Linear Algebra (2018)

- Applied Linear Algebra for Engineering (2018)
- Graph Theory (2014, 2016, 2017)
- Elements of Combinatorics (2014, 2016, 2017)

## SERVICE

- **Supervision/Mentorship at the University of Waterloo**
  - Graduate students:
    - \* Xinyue Fan (MSc, 2024 – 2026, joint with Sophie Spirkl)  
Thesis: *On induced saturation in graphs and tournaments*.
  - Directed Reading Program (DRP):
    - \* Spring 2024 project: *Non-constructive methods in combinatorics*  
Mentees: Anna Henderson and Gioia De Leonardis
    - \* Fall 2023 project: *Introduction to graph minor theory*  
Mentees: Xinyue Fan and Lyncy Li
  - Undergraduate Research Assistant Program (URA):
    - \* Spring 2023 project: *Maximum transitive set in  $H$ -free tournaments*  
Mentee: Yun Xing
- **Organizing**
  - *Themes and trends in structural graph theory* – Organizer  
Workshop at the American Institute of Mathematics (AIM), to be held in 2027
- **Refereeing for journals and conference proceedings:** IMRN, Combinatorica, JCTb, Innov. Graph Theory, Eur. J. Combin., SIDMA, J. Graph Theory, Electron. J. Combin., SODA, Euro-Comb, Workshop on Graphs.